

Assignment 2 Cloud Computing
LAUNCHING A WEB SERVER IN THE CLOUD
CS 481 Spring 2026

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1. ABSTRACT

This report documents the setup of a basic web server on Google Compute Engine using a Linux virtual machine and the Nginx web server. Multiple virtual machines were created to model a front end server, back end servers, and a remote server in a different region. Nginx was installed and basic connectivity tests were performed using internal and external network addresses. Some goals could not be fully verified because the web page from the front end external address did not load and the Nginx configuration file edits did not save as expected.

2. INTRODUCTION

Infrastructure as a Service, also called IaaS, allows a user to provision compute resources such as virtual machines on demand. This assignment focused on Google Compute Engine and the use of Nginx as a simple web server. (1, 2) The main objectives were to create and connect to Linux virtual machines, install and verify Nginx, configure network access, and document results with screenshots. Additional objectives included reverse proxying to a back end server, measuring network latency across regions, and planning for redundancy using multiple back end servers.

3. IMPLEMENTATION DETAILS

3.1. Task 1 Single VM Web Server. A front end virtual machine named web-server-1 was created in region us-central1 with a Debian based boot disk. The E2 series was selected with a small machine type to minimize cost. Firewall settings were reviewed so that HTTP traffic could reach the instance. After connecting using the browser based SSH terminal, the package list was updated and Nginx was installed.

The first attempt to run the update command included an extra line that caused a shell error. The command was then rerun correctly and installation proceeded. A simple HTML page was written to /var/www/html/index.html to replace the default Nginx content.

3.2. Task 2 Two VM Architecture. A back end virtual machine named backend-server-1 was created in the same region and network as the front end. Nginx was installed on the back end and the default page was replaced with a custom message. The internal address of the back end was recorded for use by the front end. The plan was to update the front end Nginx site configuration so that requests would be proxied to the back end using the internal address. Edits to the

Nginx configuration file did not save as expected and this prevented successful verification in the browser.

3.3. Task 3 Cloud Geography and Network Latency. A remote virtual machine named `remote-server` was created in region `asia-northeast1`. Network latency was measured from the front end server using `ping`. Tests targeted the local back end internal address, the remote internal address, and the remote external address.

3.4. Task 4 Load Balancing. A second back end named `backend-server-2` was created in the same region as the other servers but in a different zone. The goal was to configure an upstream group in the front end Nginx configuration to distribute requests across both back end servers. Because the front end page could not be loaded from the browser and Nginx configuration edits did not save as expected, full load balancing verification could not be completed.

4. RESULTS

4.1. Virtual machine inventory. Table 1 summarizes the instances and network addresses used during the lab.

Instance	Zone	Internal IP	External IP
web-server-1	us-central1-c	10.128.0.2	34.10.230.187
backend-server-1	us-central1-c	10.128.0.4	136.119.120.36
backend-server-2	us-central1-a	10.128.0.5	34.58.49.9
remote-server	asia-northeast1-b	10.146.0.2	35.221.78.29

TABLE 1. Instance names, zones, and network addresses

4.2. Screenshots and evidence.

4.3. Public browser verification. Public access was attempted by navigating to the front end external address in a browser as required by the lab instructions. The link `http://<YOUR_VM_EXTERNAL_IP>` did not load during testing, which prevented capturing the required browser screenshots. This also prevented verification of reverse proxy output and load balancing output from the same front end address.

4.4. Latency measurements. The ping tests showed low latency on the local network and higher latency when reaching the remote region. Table 2 summarizes the observed round trip times from the screenshots.

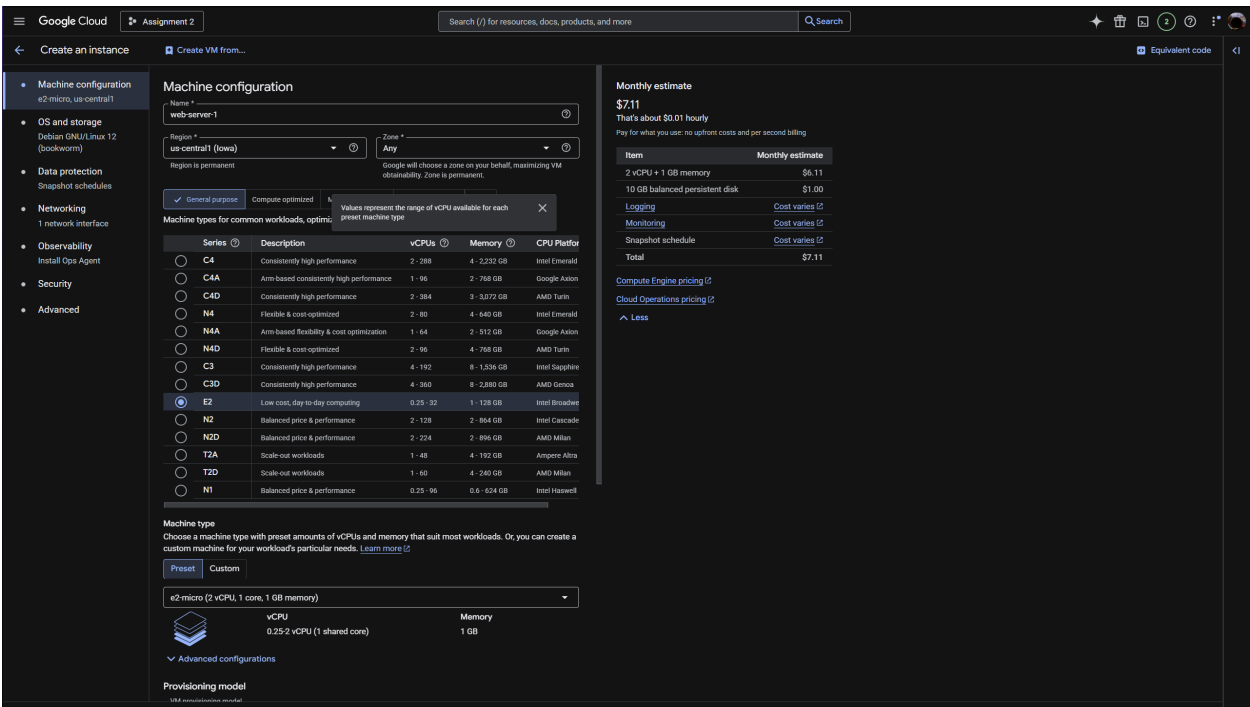


FIGURE 1. Front end instance configuration showing the E2 series selection

☐ Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	backend-server-1	us-central1-c			10.128.0.4 (nic0)	136.119.120.36 (nic0)	SSH ↓ ⋮
☐	backend-server-2	us-central1-a			10.128.0.5 (nic0)	34.58.49.9 (nic0)	SSH ↓ ⋮
☐	remote-server	asia-northeast1-b			10.146.0.2 (nic0)	35.221.78.29 (nic0)	SSH ↓ ⋮
☐	web-server-1	us-central1-c			10.128.0.2 (nic0)	34.10.230.187 (nic0)	SSH ↓ ⋮

FIGURE 3. VM instances list showing the four instances and their internal and external addresses

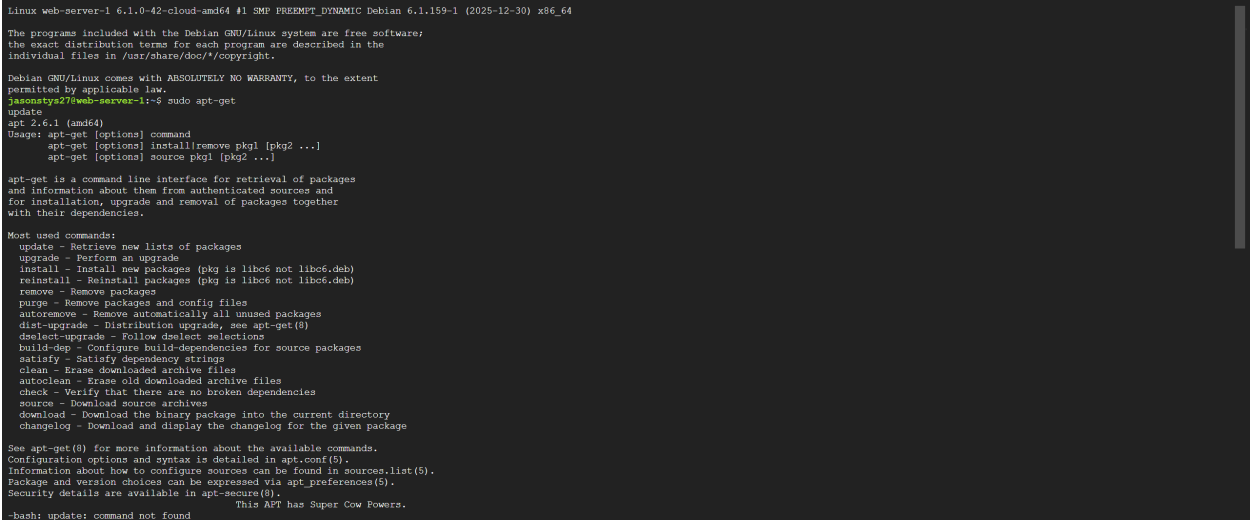


FIGURE 8. First update attempt on the front end showing a shell error

```
jasonstys27@web-server-1:~$ sudo apt-get install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 11 not upgraded.
Need to get 640 kB of archives.
After this operation, 1696 kB of additional disk space will be used.
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 https://deb.debian.org/debian bookworm/main amd64 nginx-common all 1.22.1-9+deb12u3 [112 kB]
Get:3 https://deb.debian.org/debian bookworm/main amd64 nginx amd64 1.22.1-9+deb12u3 [528 kB]
Fetched 640 kB in 0s (3641 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 74513 files and directories currently installed.)
Preparing to unpack .../nginx-common-1.22.1-9+deb12u3_all.deb ...
Unpacking nginx-common (1.22.1-9+deb12u3) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx-1.22.1-9+deb12u3_amd64.deb ...
Unpacking nginx (1.22.1-9+deb12u3) ...
Setting up nginx-common (1.22.1-9+deb12u3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /lib/systemd/system/nginx.service.
Setting up nginx (1.22.1-9+deb12u3) ...
Upgrading binary: nginx.
Processing triggers for man-db (2.11.2-2) ...
```

FIGURE 9. Nginx installation on the front end

```
jasonstys27@web-server-1:~$ sudo systemctl
status nginx
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
proc-sys-fs-binfmt_misc.automount  loaded active running Arbitrary Executable File Formats File System Automount Point
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda1.device loaded active plugged PersistentDisk 1
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda14.device loaded active plugged PersistentDisk 14
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda15.device loaded active plugged PersistentDisk 15
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda15.device loaded active plugged PersistentDisk
sys-devices-pci0000:00-0000:00:04-0-virtio0-net-ens4.device loaded active plugged Virtio network device
sys-devices-pnp0-00:03-tty-tyt50.device loaded active plugged /sys/devices/pnp0/00:03/tty/tyt50
sys-devices-pnp0-00:04-tty-tyt51.device loaded active plugged /sys/devices/pnp0/00:04/tty/tyt51
sys-devices-pnp0-00:05-tty-tyt52.device loaded active plugged /sys/devices/pnp0/00:05/tty/tyt52
sys-devices-pnp0-00:06-tty-tyt53.device loaded active plugged /sys/devices/pnp0/00:06/tty/tyt53
sys-module-configfs.device loaded active plugged /sys/module/configfs
sys-module-fuse.device loaded active plugged /sys/module/fuse
sys-subsystem-net-devices-ens4.device loaded active plugged Virtio network device
~.mount loaded active mounted Root Mount
boot-efi.mount loaded active mounted /boot/efi
dev-hugepages.mount loaded active mounted Huge Pages File System
dev-mqueue.mount loaded active mounted POSIX Message Queue File System
proc-sys-fs-binfmt_misc.mount loaded active mounted Arbitrary Executable File Formats File System
run-credentials-systemd-x2dayact1.service.mount loaded active mounted /run/credentials/systemd-x2dayact1.service
run-credentials-systemd-x2dtmpfiles-x2dsetup.service.mount loaded active mounted /run/credentials/systemd-x2dtmpfiles-x2dsetup.service
run-credentials-systemd-x2dtmpfiles-x2dsetup-x2ddev.service.mount loaded active mounted /run/credentials/systemd-x2dtmpfiles-x2dsetup-x2ddev.service
run-user-1000.mount loaded active mounted
sys-fs-fuse-connections.mount loaded active plugged
sys-kernel-config.mount loaded active plugged
sys-kernel-debug.mount loaded active plugged
sys-kernel-tracing.mount loaded active plugged
systemd-ask-password-console.path loaded active plugged
systemd-ask-password-wall.path loaded active plugged
init.scope loaded active plugged
session-1.scope loaded active plugged
[Press q to quit, ^? to toggle help, enter to top, enter to bottom, ~ to top/bottom]
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
proc-sys-fs-binfmt_misc.automount  loaded active running Arbitrary Executable File Formats File System Automount Point
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda1.device loaded active plugged PersistentDisk 1
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda14.device loaded active plugged PersistentDisk 14
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda15.device loaded active plugged PersistentDisk 15
sys-devices-pci0000:00-0000:00:03-0-virtio0-host0-target0:0:1-0:0:1:0-block-sda-sda15.device loaded active plugged PersistentDisk
sys-devices-pci0000:00-0000:00:04-0-virtio0-net-ens4.device loaded active plugged Virtio network device
sys-devices-pnp0-00:03-tty-tyt50.device loaded active plugged /sys/devices/pnp0/00:03/tty/tyt50
sys-devices-pnp0-00:04-tty-tyt51.device loaded active plugged /sys/devices/pnp0/00:04/tty/tyt51
sys-devices-pnp0-00:05-tty-tyt52.device loaded active plugged /sys/devices/pnp0/00:05/tty/tyt52
sys-devices-pnp0-00:06-tty-tyt53.device loaded active plugged /sys/devices/pnp0/00:06/tty/tyt53
sys-module-configfs.device loaded active plugged /sys/module/configfs
sys-module-fuse.device loaded active plugged /sys/module/fuse
sys-subsystem-net-devices-ens4.device loaded active plugged Virtio network device
~.mount loaded active mounted Root Mount
boot-efi.mount loaded active mounted /boot/efi
dev-hugepages.mount loaded active mounted Huge Pages File System
dev-mqueue.mount loaded active mounted POSIX Message Queue File System
proc-sys-fs-binfmt_misc.mount loaded active mounted Arbitrary Executable File Formats File System
run-credentials-systemd-x2dayact1.service.mount loaded active mounted /run/credentials/systemd-x2dayact1.service
run-credentials-systemd-x2dtmpfiles-x2dsetup.service.mount loaded active mounted /run/credentials/systemd-x2dtmpfiles-x2dsetup.service
run-credentials-systemd-x2dtmpfiles-x2dsetup-x2ddev.service.mount loaded active mounted /run/credentials/systemd-x2dtmpfiles-x2dsetup-x2ddev.service
```

FIGURE 10. Systemctl output on the front end showing an incorrect command entry

```
jasonstys27@web-server-1:~$ echo "Hello from my GCE web server!
My name is [Your Name]." | sudo tee /var/www/html/index.html
Hello from my GCE web server!
My name is [Your Name].
jasonstys27@web-server-1:~$
```

FIGURE 11. Custom index.html written on the front end

FIGURE 12. Back end setup showing package update, install, and initial file path issue

FIGURE 13. Nginx service status on the back end

Nginx installation succeeded on the front end and back end as shown in Figures 9 and 13. Local connectivity between the front end and back end over the internal network was successful as shown in Figure 17. The latency to the remote server was much higher, which aligns with the

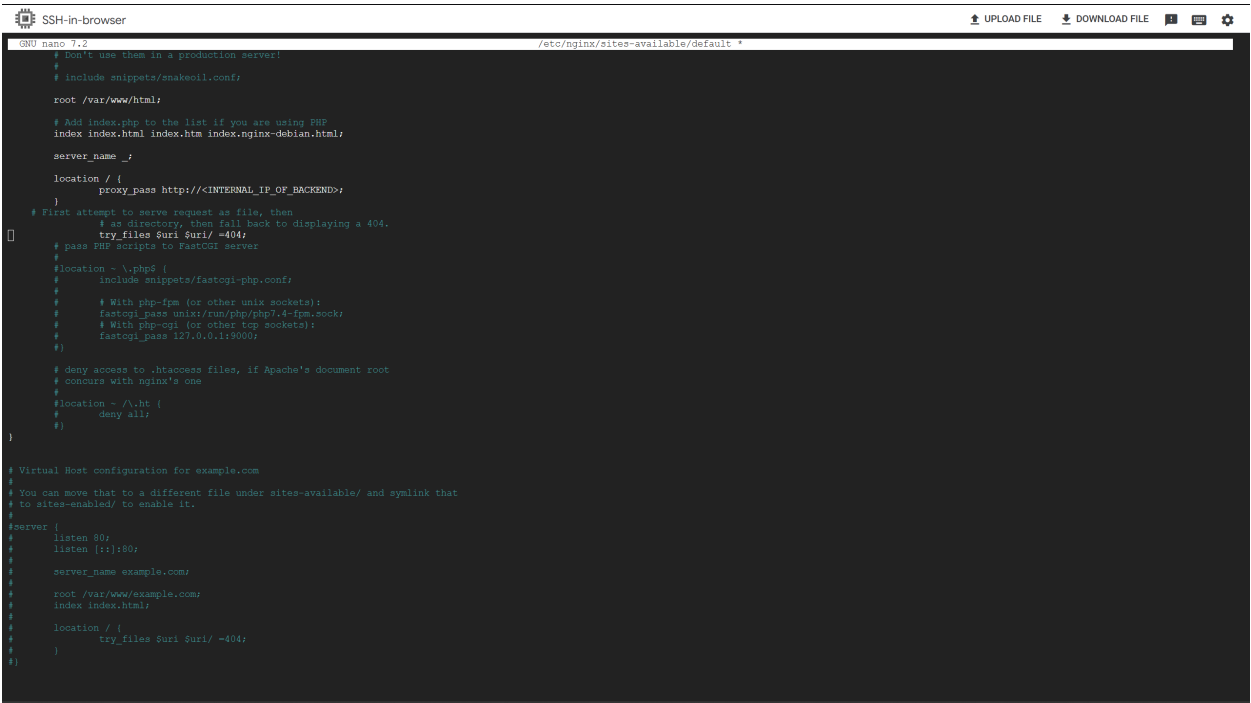


FIGURE 14. Front end Nginx site configuration opened for reverse proxy edits

Placeholder for a full screen browser screenshot that shows the front end URL and the response page. If the page does not load, capture the browser error page and include it here.

FIGURE 15. Browser evidence placeholder for the front end public address test

Placeholder for two browser screenshots that show the same front end URL while alternating between the two back end messages.

FIGURE 16. Browser evidence placeholder for load balancing verification

Destination	Observed round trip time
Local back end internal address 10.128.0.4	2.39 ms
Remote internal address 10.146.0.2	133 ms
Remote external address 35.221.78.29	146 ms

TABLE 2. Observed ping times from the front end instance

physical distance between regions and the time needed for packets to travel over long distance fiber paths.

Public browser verification was not successful during this run. The page at the front end external address did not load, so screenshots of the Nginx welcome page and the custom page in a

```
jasonstys27@web-server-1:~$ ping 10.128.0.4
PING 10.128.0.4 (10.128.0.4) 56(84) bytes of data.
64 bytes from 10.128.0.4: icmp seq=1 ttl=64 time=2.39 ms
```

FIGURE 17. Ping from front end to the local back end internal address

```
jasonstys27@web-server-1:~$ ping 10.146.0.2
PING 10.146.0.2 (10.146.0.2) 56(84) bytes of data.
64 bytes from 10.146.0.2: icmp seq=1 ttl=64 time=133 ms
```

FIGURE 18. Ping from front end to the remote internal address

```
jasonstys27@web-server-1:~$ ping 35.221.78.29
PING 35.221.78.29 (35.221.78.29) 56(84) bytes of data.
64 bytes from 35.221.78.29: icmp seq=1 ttl=52 time=146 ms
```

FIGURE 19. Ping from front end to the remote external address

browser could not be captured. This outcome most likely indicates a networking or firewall configuration issue. If HTTP traffic is not allowed to port 80 for the instance network tag, the VM can run Nginx while still being unreachable from the public internet. A useful check is to run `curl localhost` on the VM to confirm local service output, then confirm a firewall rule exists for TCP port 80 with the correct target tags.

Reverse proxy and load balancing configuration also could not be fully verified. Edits to the Nginx site configuration did not save as expected in the editor, which prevented a clean restart and browser testing. A practical approach is to ensure the file is edited using `sudo nano`, validate syntax with `sudo nginx -t`, then restart using `sudo systemctl restart nginx`. When the reverse proxy is in place, it allows the front end to accept public traffic while forwarding requests to a back end using an internal address. This separation improves security because the back end does not need a public firewall rule and it reduces exposure to the internet. It also improves maintainability because each layer can be updated independently.

Using internal addresses for service to service communication is beneficial because it avoids public routing, reduces latency, and limits access to the private network. Scaling to multiple back ends introduces additional challenges. Health checks and monitoring are needed to remove unhealthy servers from rotation. Logging and metrics need aggregation across instances. Configuration management becomes important to keep Nginx settings consistent across changes.

6. CONCLUSION

Provided practice with creating and managing Linux virtual machines in Google Compute Engine and installing Nginx as a web server. The multi VM setup also demonstrated internal networking and the effect of geographic distance on latency. Some tasks could not be completed fully due to issues reaching the front end from a browser and issues saving Nginx configuration edits. The troubleshooting steps identified in the discussion section provide a clear path to complete public access, reverse proxying, and load balancing in a follow up run.

REFERENCES

1. Google Cloud, *Compute Engine documentation* (2026; <https://cloud.google.com/compute/docs>).
2. NGINX, *NGINX documentation* (2026; <https://nginx.org/en/docs/>).

APPENDIX A. COMMAND LOG

The list below provides a quick reference of commands used during the lab.

- (1) `sudo apt-get update`
- (2) `sudo apt-get install -y nginx`
- (3) `sudo systemctl status nginx`
- (4) `echo "Hello from my GCE web server! My name is Jason Stys." | sudo tee /var/www/html/index.html`
- (5) `sudo nano /etc/nginx/sites-available/default`
- (6) `sudo nginx -t`
- (7) `sudo systemctl restart nginx`
- (8) `ping 10.128.0.4`
- (9) `ping 10.146.0.2`
- (10) `ping 35.221.78.29`